Cell organelles 3

Dr. Iman Nabil

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Why?
tell me
why!



Cell organelles 3

1- Mitochondria.

2- Peroxisome.

a. Structure.

- LM
- EM

b. Sites.

c. Function.

Dr Iman Nabil

I-Mitochondria

Dr Imad Sabbil

A-Function of Mitochondria

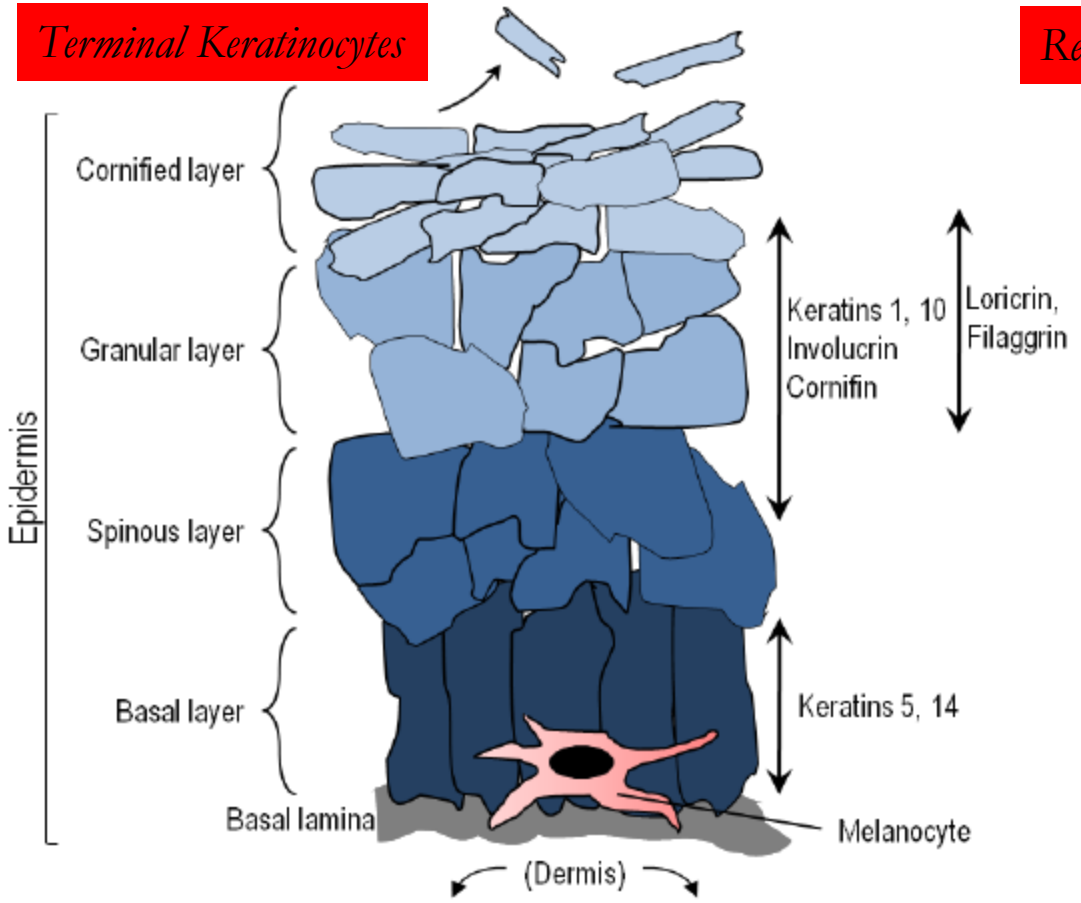
- Powerhouses of the cell as they are the sites of adenosine triphosphate (ATP) production.

Dr Iman Nabil

B-Sites of Mitochondria

- All cells Except:

Terminal Keratinocytes

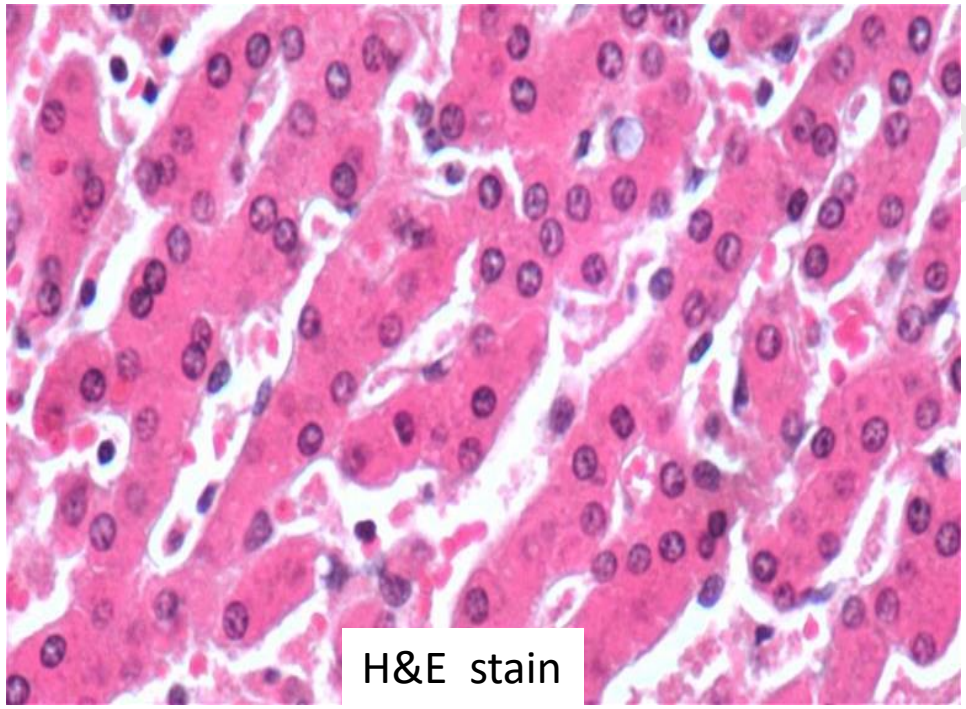


Red blood corpuscles

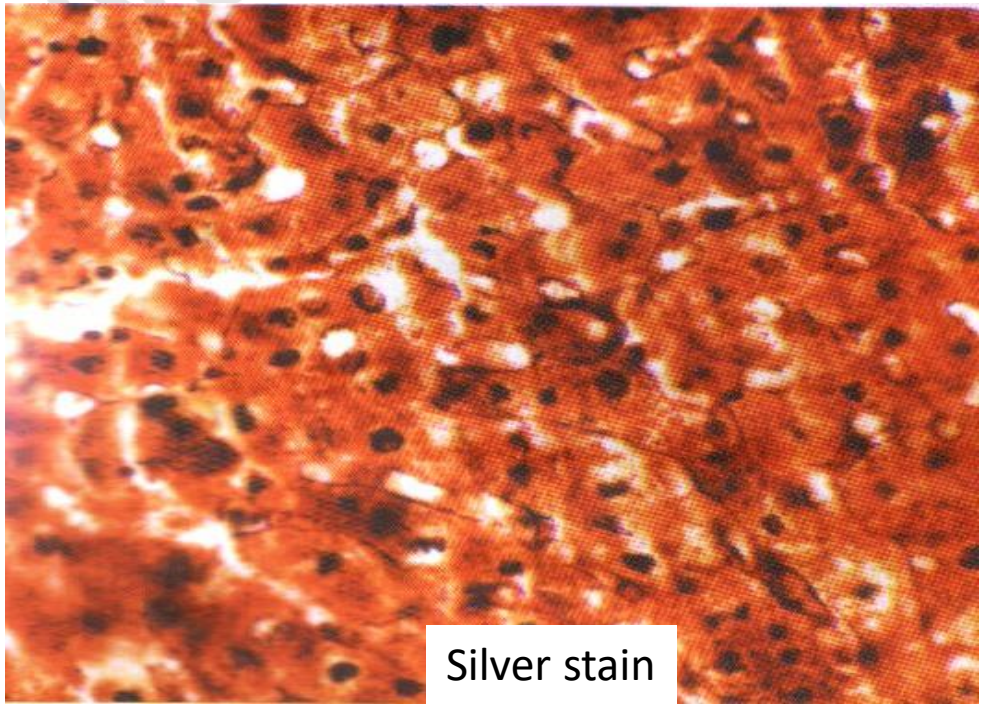


C- Structure
1-LM of mitochondria

- When present in large numbers contribute to *the cytoplasmic eosinophilia*.



H&E stain



Silver stain

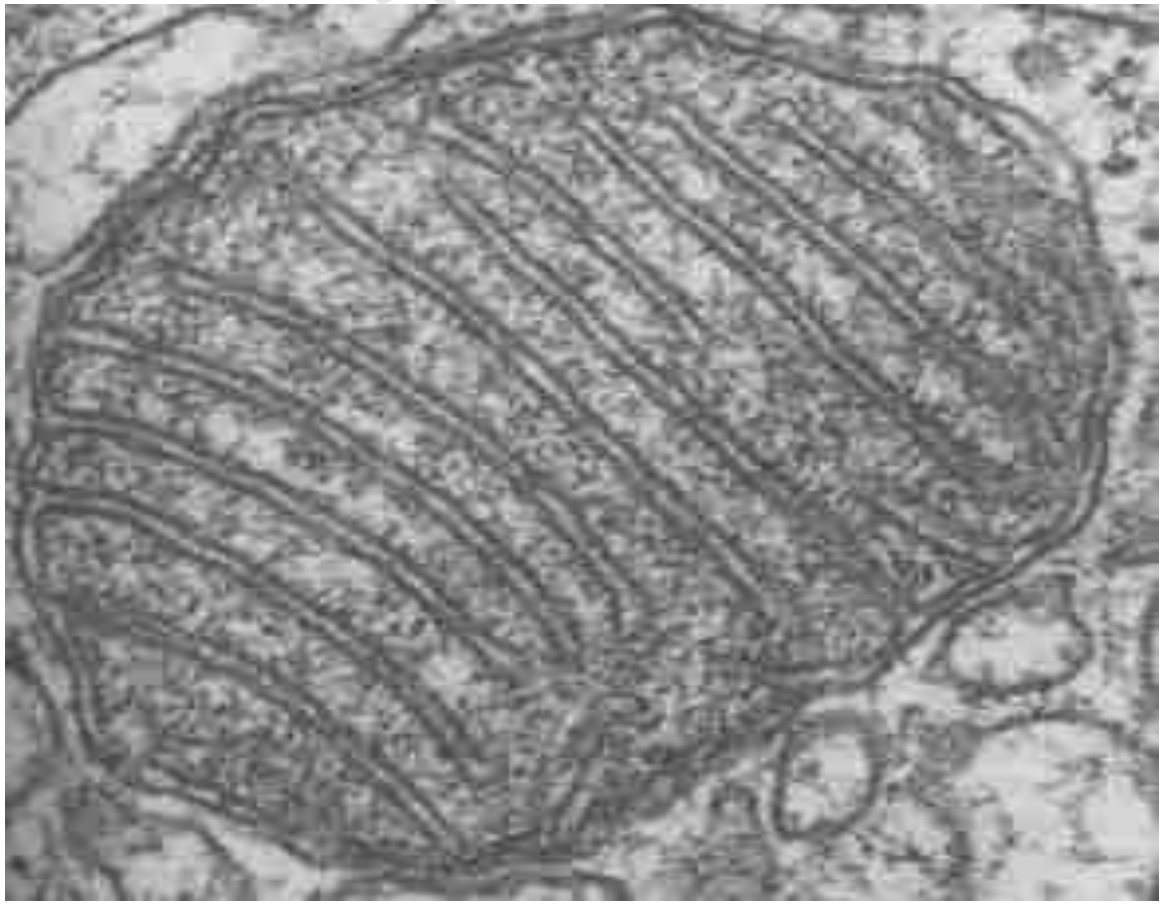
Nabil

C- Structure
2-EM of mitochondria

1- Membrane-bounded organelles.

2- Surrounded by *two membranes: outer and inner*, which define two mitochondrial compartments:

- The intermembranous space: between the two membranes.
- The matrix space: enclosed by the inner membrane.

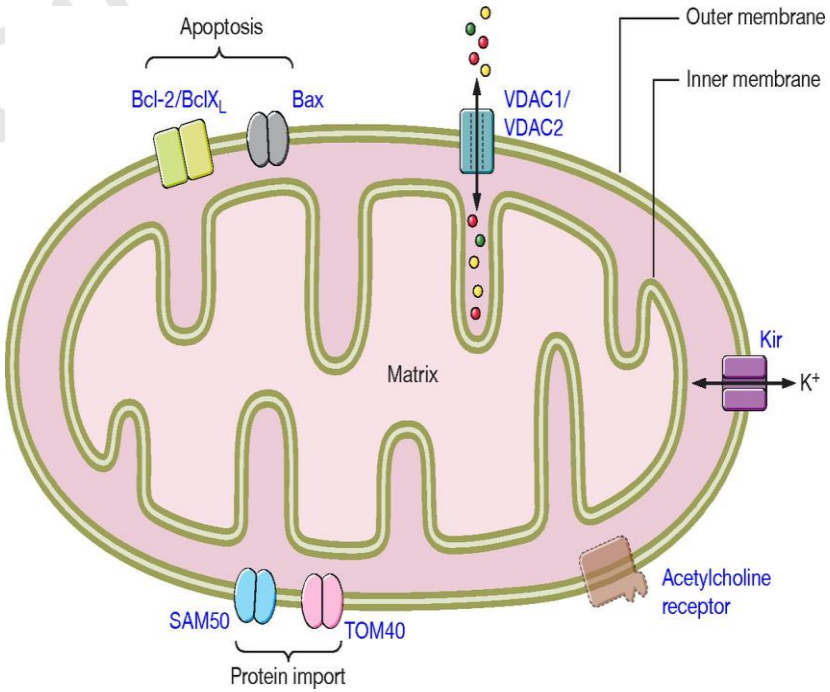


Dr Imar

1-The outer mitochondrial membrane

- It is *smooth* and *porous*.

Function: allows passage of small molecules due to the presence of specific transmembrane proteins called *porins*.

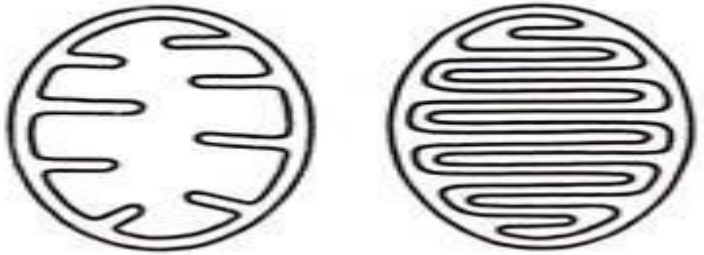


Dr Iman Nar :)

2- The inner mitochondrial membrane

1- Folded into *cristae* which increase its surface area; the number of cristae is greater in cells of greater demand for ATP.

Shelf/lamellar-like cristae Tubular cristae



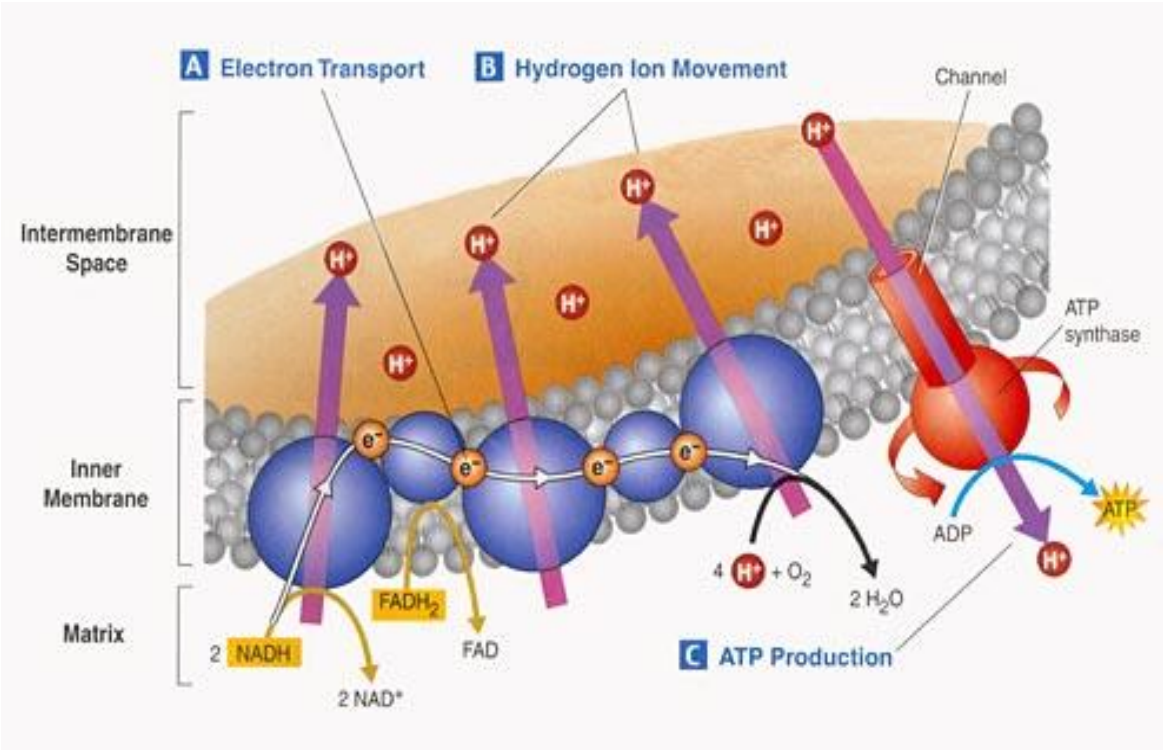
- Types of cristae:

➢ lamellar cristae : most of cells.

➢ tubular cristae: steroid secreting cells.

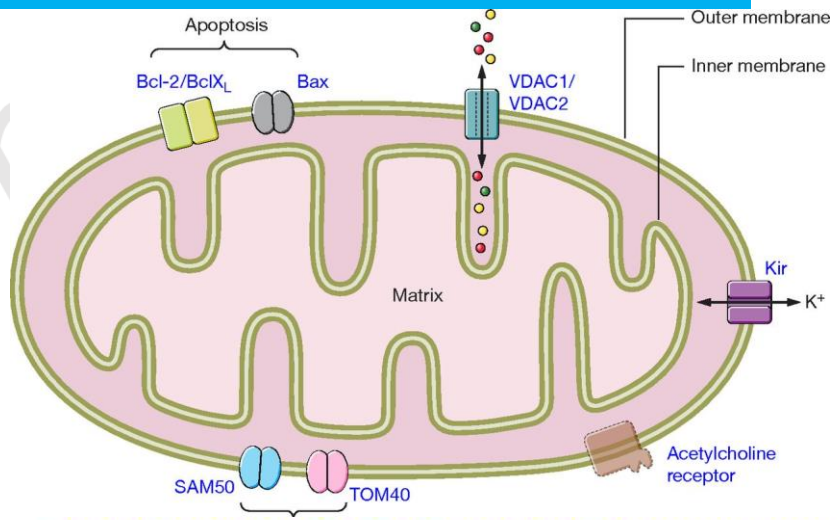
2- *Impermeable* to ions and small molecules due to presence of phospholipid called *cardiolipin*.

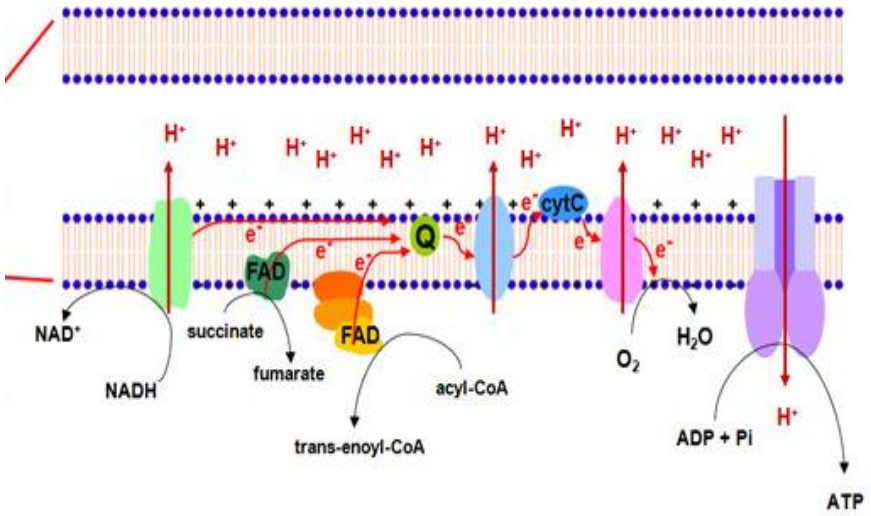
Function: contains the enzymes of the electron transport system (respiratory chain enzymes) and the ATP synthase.



3- Intermembranous space

- Contains substances *diffusing* from the cytoplasm through the outer membrane and ions *pumped out* of the matrix space through the inner membrane.





Dr Iman Nah

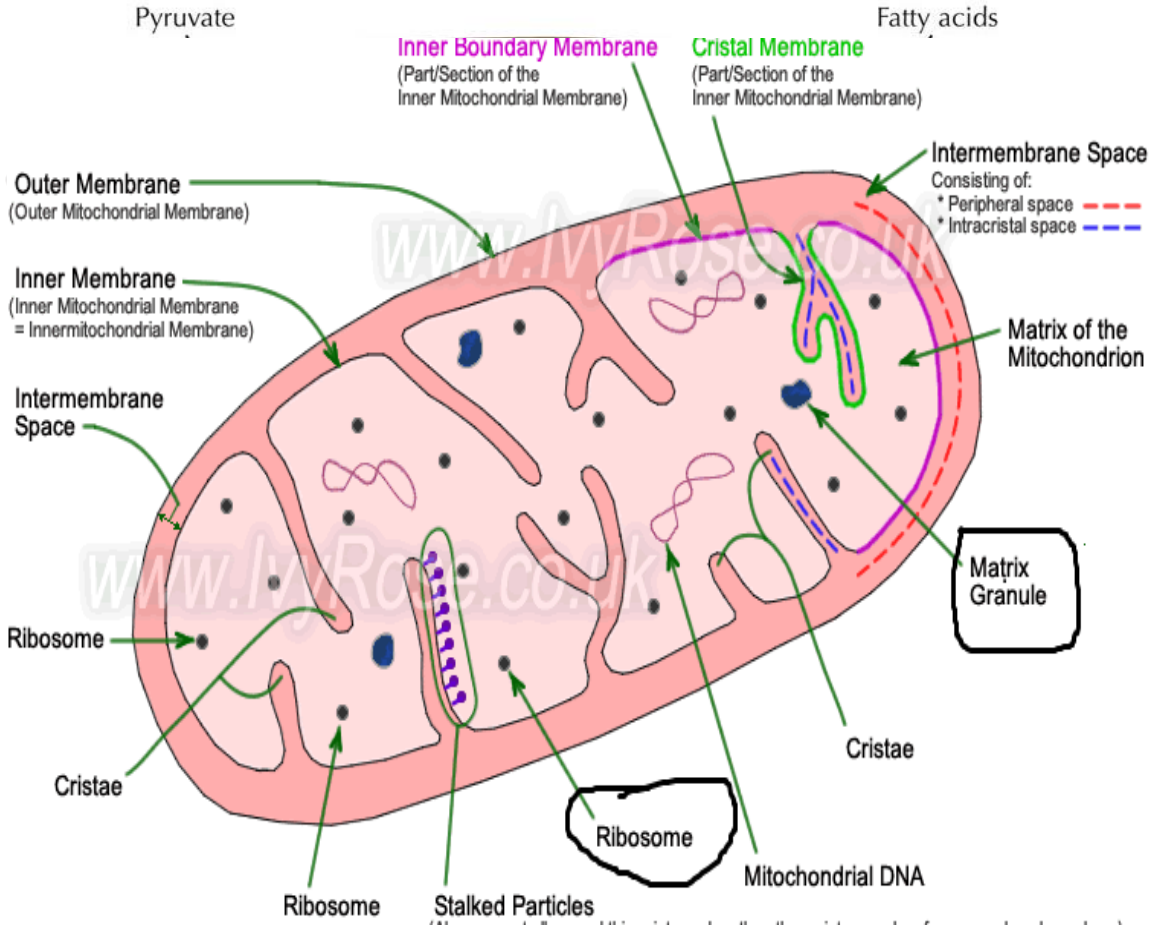
4-Matrix space

- Surrounded by the inner mitochondrial membrane.
- Functions:

1- *Enzymes* involved in mitochondrial functions as citric acid cycle.

2- Mitochondrial *DNA* and few *ribosomes*.

3- *Matrix granules:* store calcium ions, play a role in mitochondrial regulation of Ca^{2+} intracellular concentration.



The genetic system of mitochondria

- *The mitochondrial DNA:*

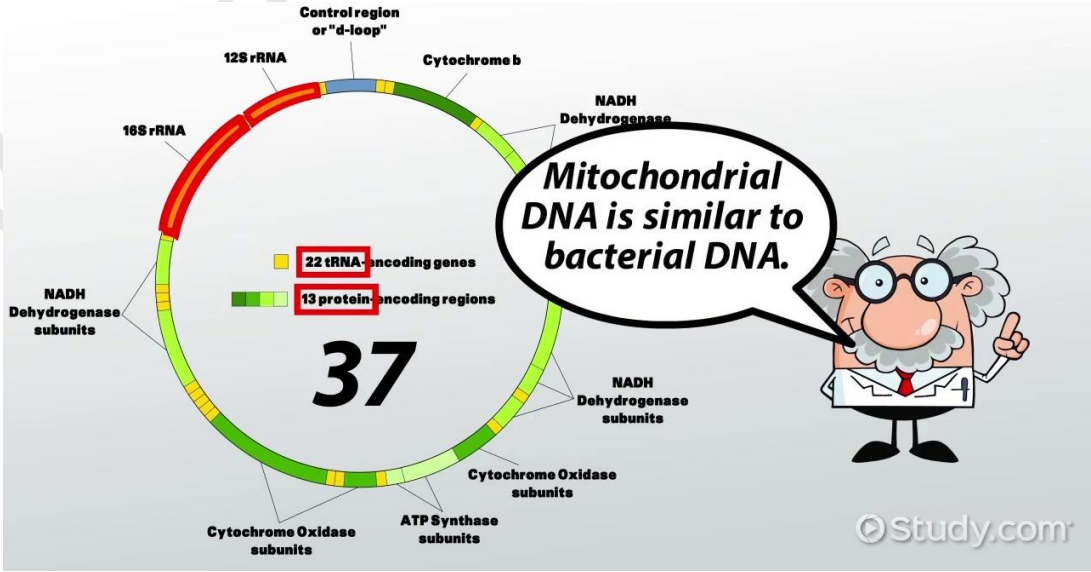
1- a circular molecule.

2- limited coding capacity.

3- represents 1% of the total DNA of the cell.

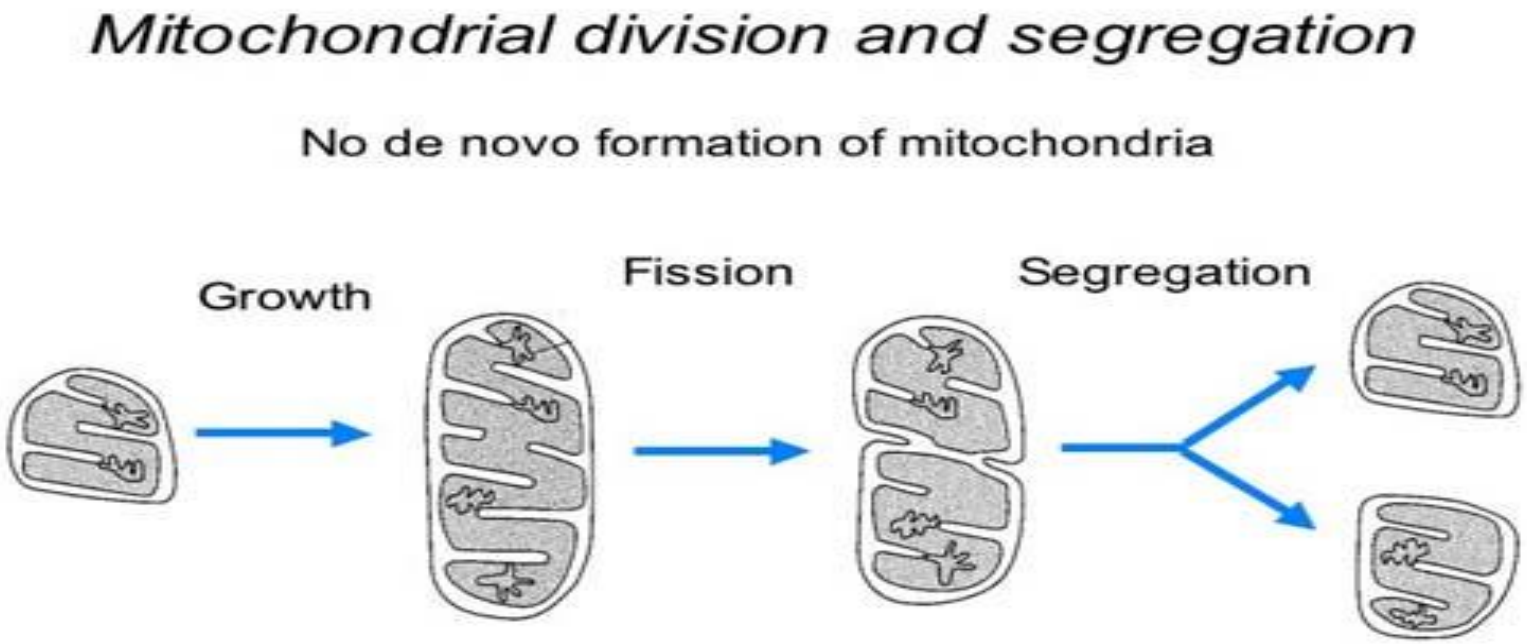
Function: Mitochondria can synthesize some of their structural proteins by their own RNAs.

- *Most of the mitochondrial proteins are encoded by the nuclear DNA and are synthesized in the cytoplasm and imported into mitochondria.*



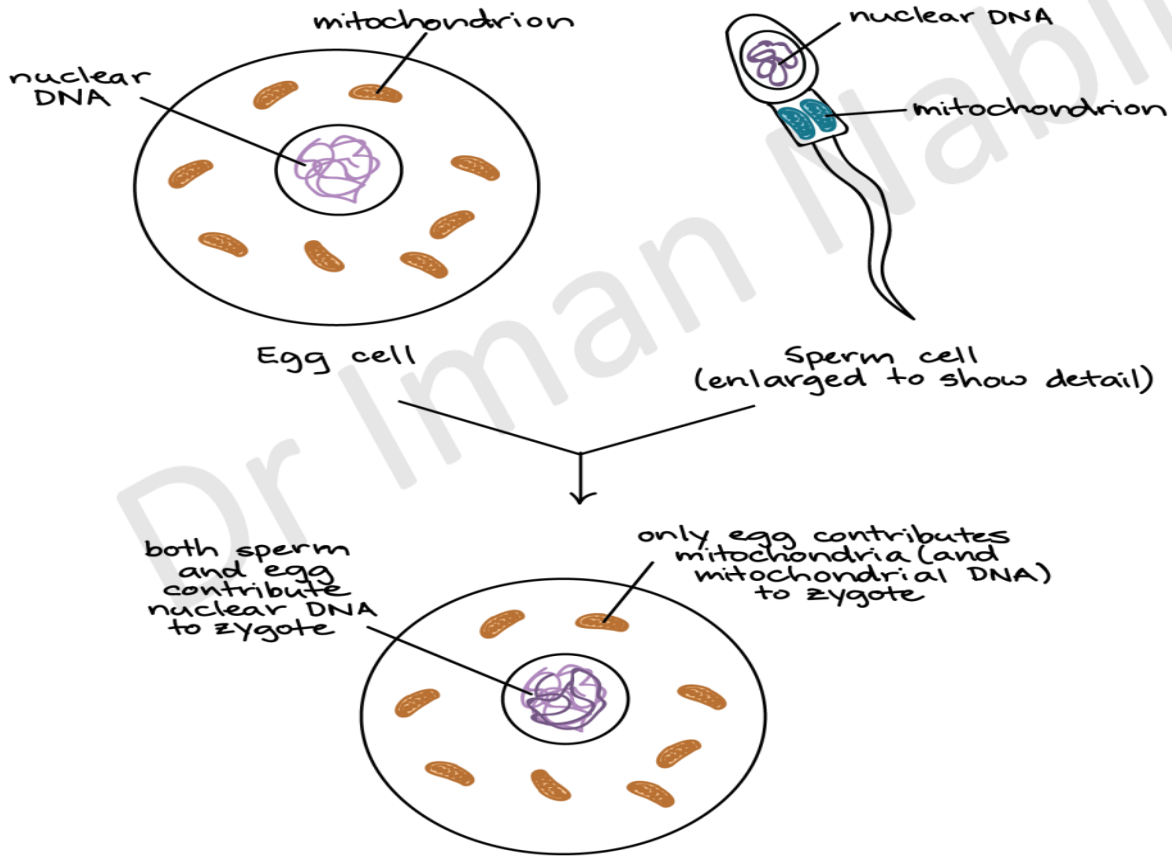
The genetic system of mitochondria

- Mitochondria are self-replicating organelles.



What is your source of mitochondria?

Dr Imran Sohil



Dr Iman Nabil

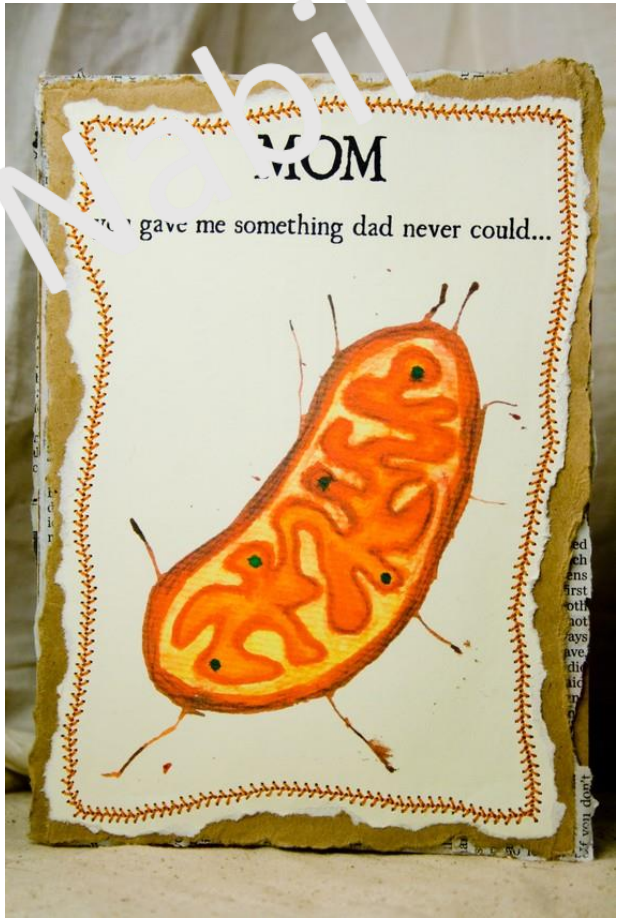


Thanks for the Mito*, Mom!

HAPPY MOTHER'S DAY

FROM 23andMe

* Mito is short for "mitochondria," which are small structures inside of cells that have their own DNA. Since children (both boys and girls) only inherit their mitochondrial DNA from their mother, it can be used to trace a person's maternal line.



Dr Iman Mabil

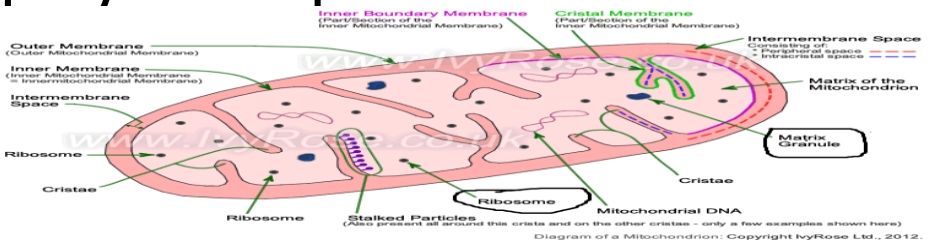
How do mitochondria adapt to its function?

1- Outer membrane: smooth &porous contains mitochondrial porins → allow easy passage of small molecules.

2- Inner membrane is folded into numerous cristae → increase surface area for energy production.

3- Inner membrane contains cardiolipin → make it highly impermeable to ions & small molecules.

4- Matrix space: contains enzymes for citric acid cycle, mito DNA & ribosomes → synthesize some of their structural proteins, also contains matrix granules → store Ca thus play an important role in regulation of intracellular Ca concentration.



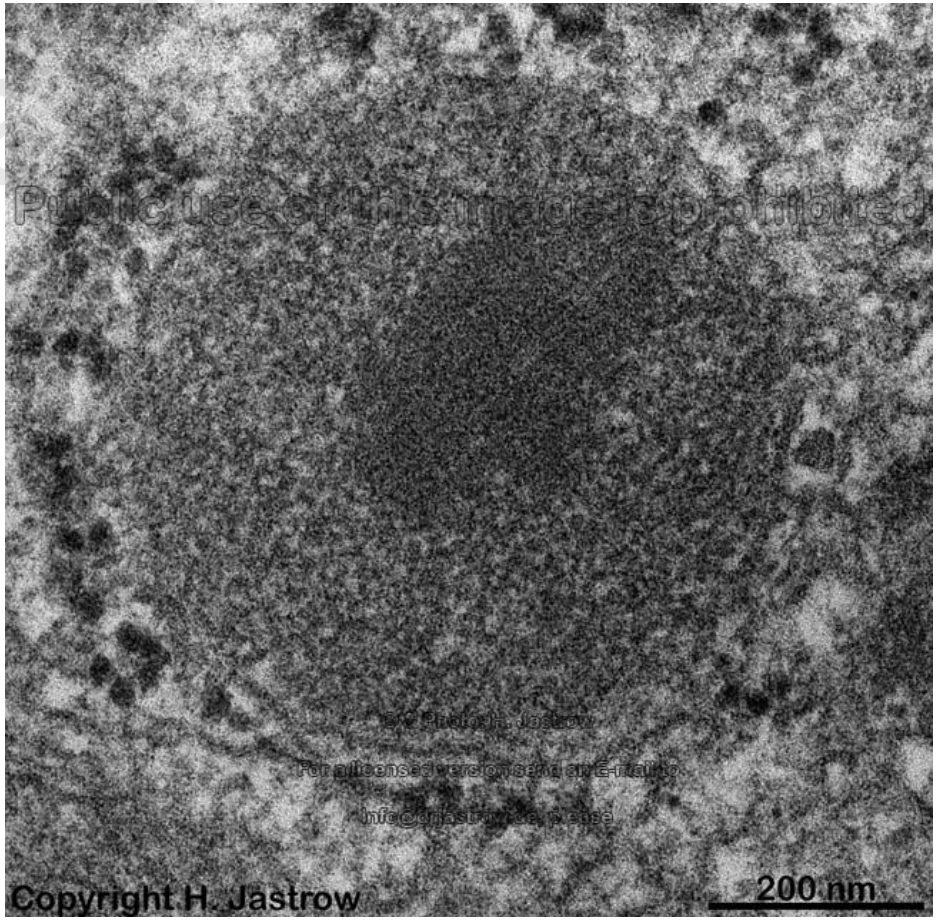
II-Peroxisome

Dr Imad Kabbil

Peroxisome

Definition: membrane-bounded organelles that contain oxidative enzymes.

- Peroxisomes possess *no genetic material* of their own.



Dr Iman Nabil

A-Structure of Peroxisome

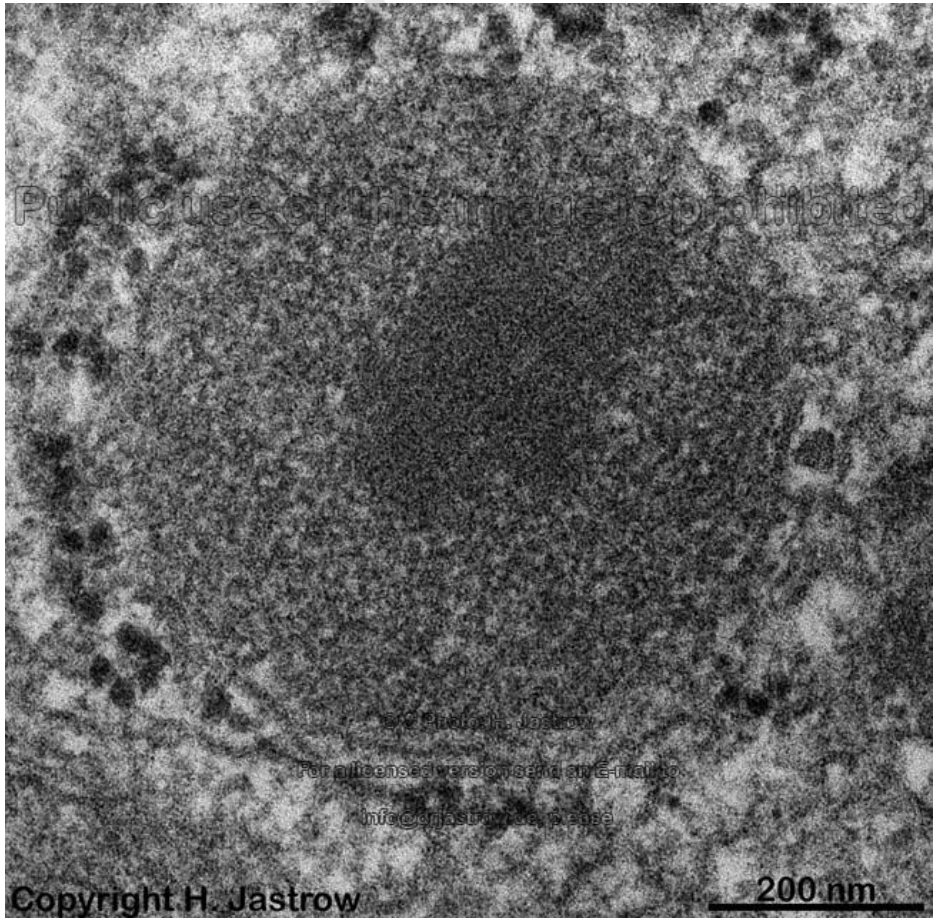
- **LM:**

They are not seen by H&E stain.

- **EM:**

1- Small, spherical bodies with fine granular electron dense content.

2- Surrounded by a single membrane.



Dr Iman

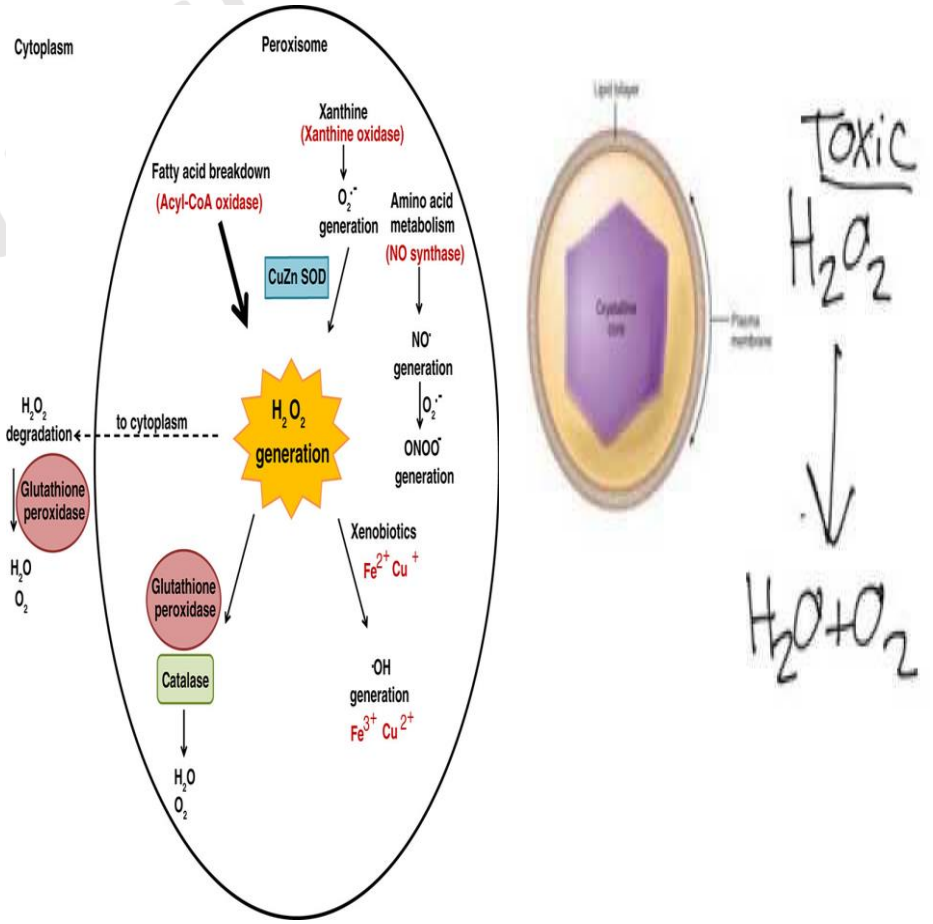
B-Functions of peroxisomes

1- *β-oxidation of long chain fatty acids to release energy*. However, they differ from mitochondria in that they are unable to store this energy in the form of ATP. This energy is released as *heat* to maintain body temperature.

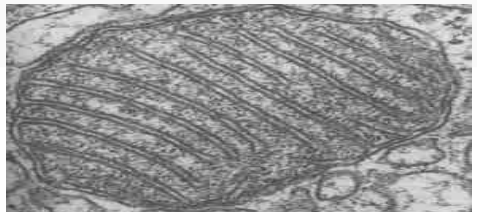
2- *Generation of hydrogen peroxide*, which detoxifies toxic agents.

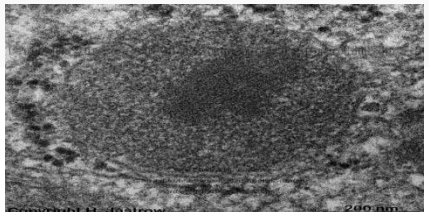
3- Contain *catalase enzyme* that converts the excess hydrogen peroxide into water, thus protecting the cell.

4- *Detoxification of alcohol* in cooperation with the smooth endoplasmic reticulum in the liver.



	Mitochondria	Peroxisome
1- LM	-Not seen by H&E except in large amount cause cytoplasmic acidophilia. -By special stain (silver stain) appear as brownish granules.	Not seen.
2- EM	Double membrane: outer is smooth and inner is folded into cristae enclosed matrix space.	Single membrane enclosed fine granular contents.
3- Function	Production of energy & store it in the form of ATP	1- Produce energy & released it in the form of heat (unable to store it). 2- Produce hydrogen peroxide. 3- convert excess hydrogen peroxide into water. 4- Detoxification of toxic substances.
4- sites	All body cells except red blood cells & keratinocytes.	Many cells especially liver.
5- Genetic material	Present	Absent





Dr Iman Nabil

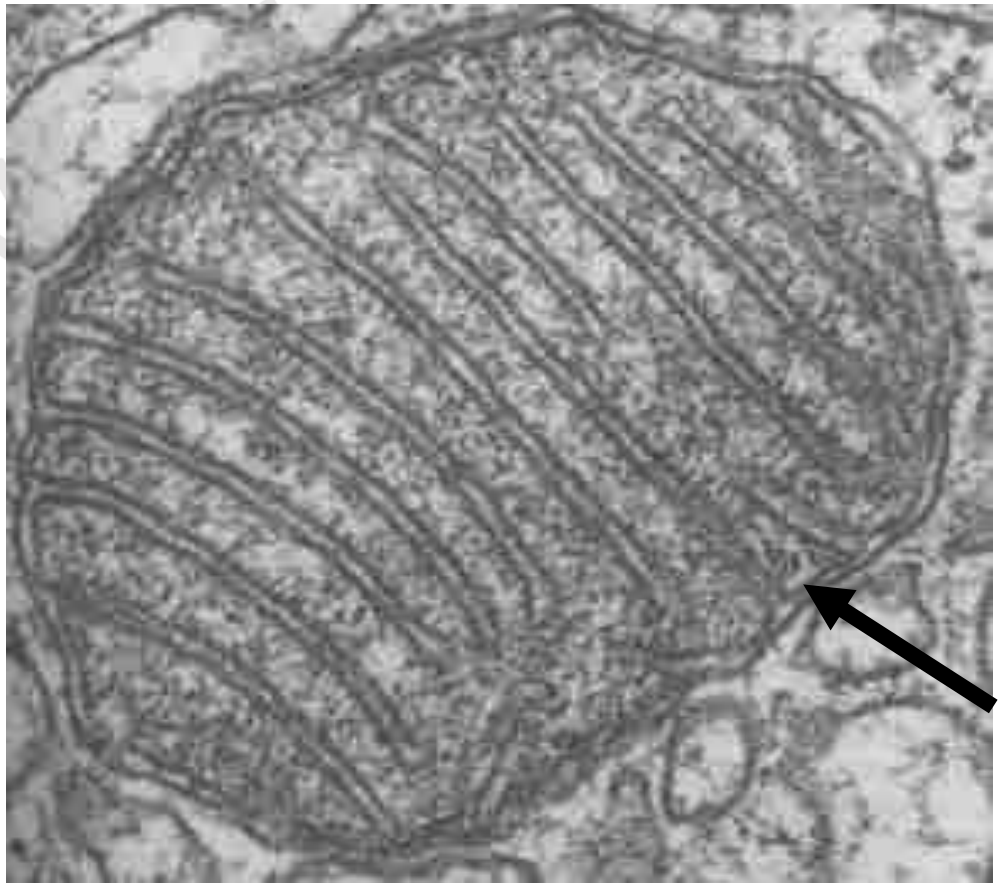
Which statement is the best to describe the pointed structure?

1- It contains porins that allow the passage of small molecules.

2- It contains the enzymes of citric acid cycle.

3- It is freely permeable to H ions.

4- It contains cardiolipin.



Dr Iman N

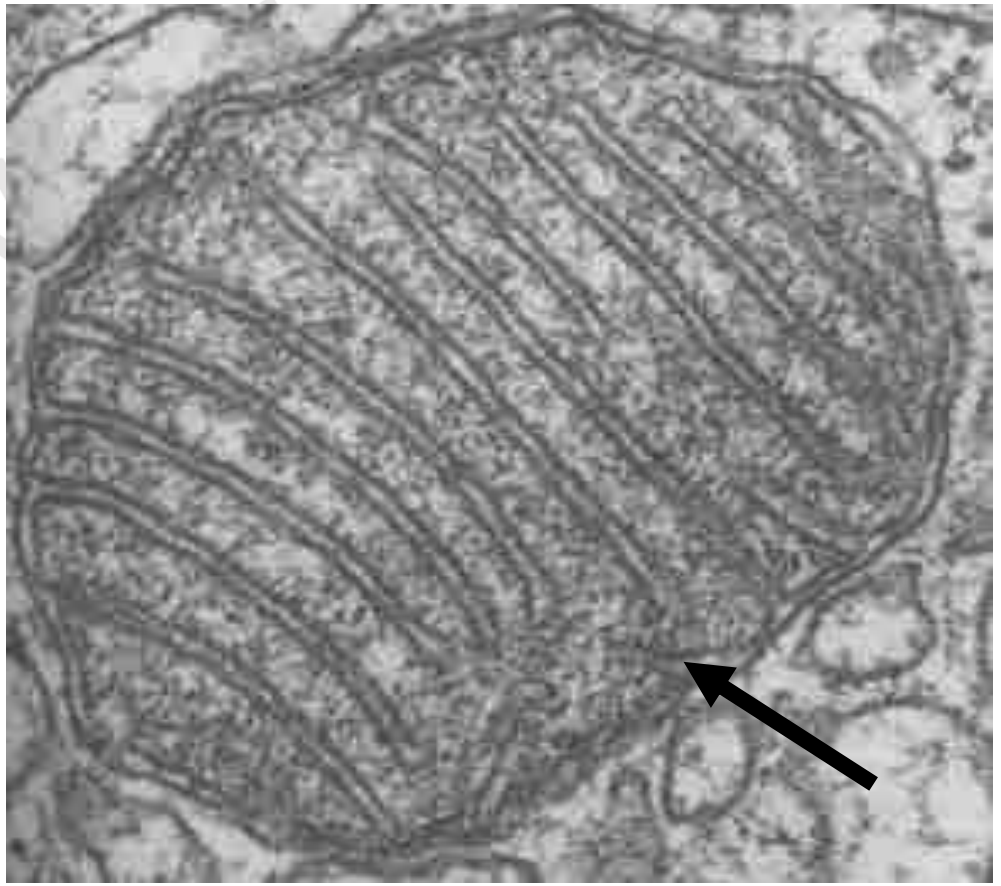
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Dr Iman N


THANK
YOU


Thank You